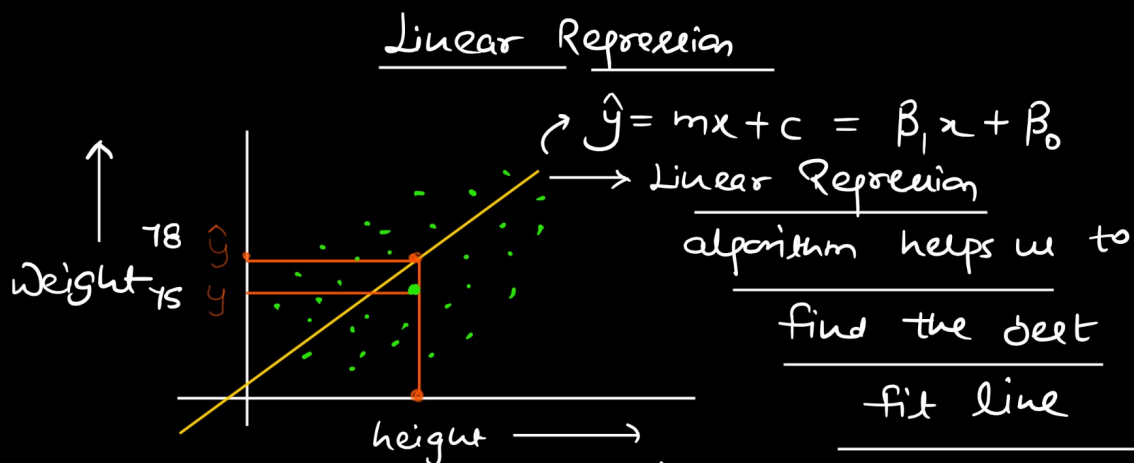


Simple Linear Regression



① How to get this

best fit line

$\propto m \& c$

$\hookrightarrow$ best/optimized value of $m \& c$

$x \rightarrow$ independent feature

$m \rightarrow$ slope/coefficient

$c \rightarrow$ intercept

$\hat{y} \rightarrow$ prediction

Ideal goal

Error $\Rightarrow (y - \hat{y})$ $\downarrow \downarrow$
Residuals

$\hookrightarrow$ to reduce the error

Real time scenario

(Multiple Linear Regression)

$$\hat{y} = m_1 x_1 + m_2 x_2 + m_3 x_3 + m_4 x_4 + c$$

$\propto (x_1, x_2, x_3, x_4) \rightarrow$ input/
independent
features

$\propto (m_1, m_2, m_3, m_4) \rightarrow$ slopes

MSE (Mean Squared Error)

$$\Rightarrow \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n}$$

$n \rightarrow$ num of data points

MAE (Mean Absolute Error)

outliers
in your
data

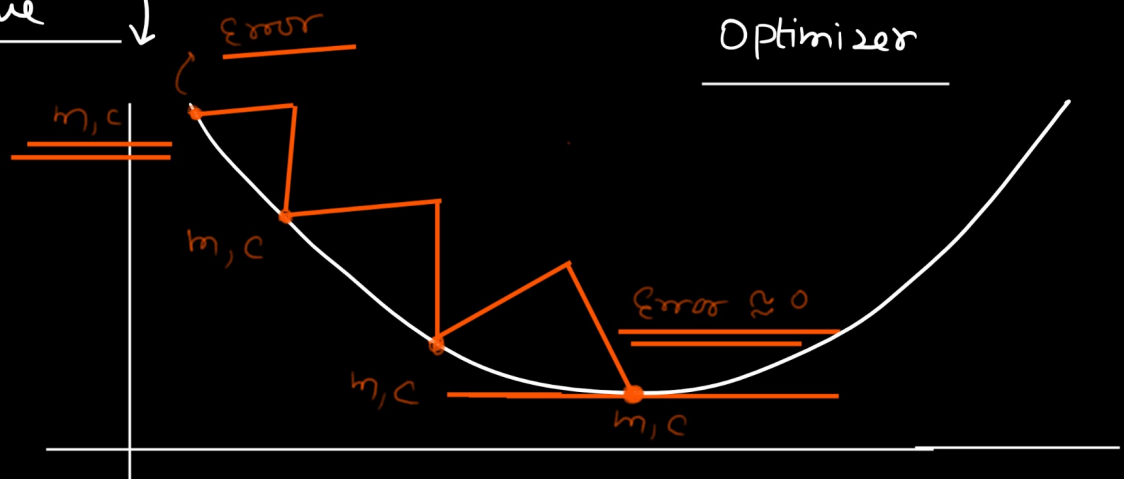
$$\Rightarrow \frac{\sum_{i=1}^n |y_i - \hat{y}_i|}{n}$$

How to get the best value of m & c ?

↳ Optimizers

Convex
curve

↳ Gradient Descent
Optimizer



Gradient Descent

Optimizers

$$m_{\text{new}} = m_{\text{old}} - \eta * \frac{\partial L}{\partial m_{\text{old}}}$$

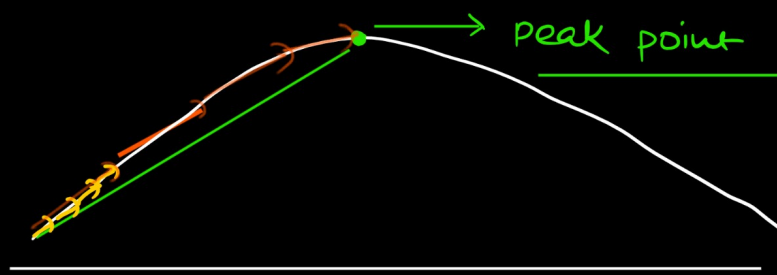
$$c_{\text{new}} = c_{\text{old}} - \eta * \frac{\partial L}{\partial c_{\text{old}}}$$

Learning Rate ≈ 0.001

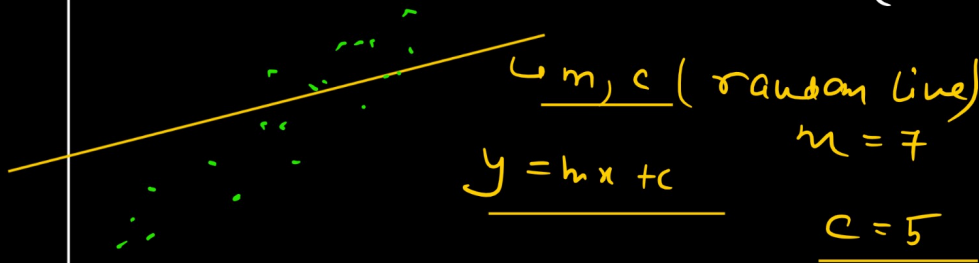
(Hyperparameter)

Loss function

↳ Error (MSE)



Cost function $\rightarrow \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$



$c = 5$

Loss function /
Cost function /

Error /

Residuals

$m_{new} = 7 - 0.001 \times \frac{\delta L}{\delta m}$

$c_{new} = 5 - 0.001 \times \frac{\delta L}{\delta c}$

$\frac{\delta x^2}{\delta x} = \underline{\underline{2x}}$

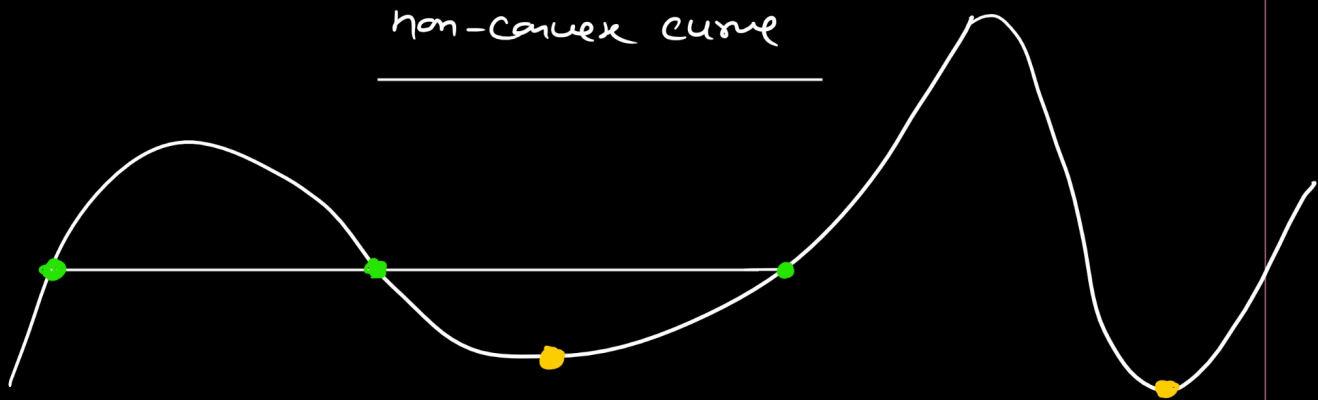
$\frac{\delta L}{\delta m} = \frac{1}{n} \frac{\delta \sum_{i=1}^n (y_i - \hat{y}_i)^2}{\delta m}$

$= \frac{1}{n} \frac{\delta \sum_{i=1}^n (y_i - (mx_i + c))^2}{\delta m}$

$= \frac{1}{n} \sum_{i=1}^n 2 (y_i - (mx_i + c)) \times \frac{\delta (y_i - (mx_i + c))}{\delta m}$

$\frac{\delta L}{\delta m} \Rightarrow \frac{1}{n} \sum_{i=1}^n 2 (y_i - (mx_i + c)) \times (-x_i)$

$$\frac{\partial L}{\partial \eta} \Rightarrow -\frac{2}{\eta} \sum_{i=1}^{\eta} x_i (y_i - \hat{y}_i)$$



$$\frac{\partial L}{\partial c} = \frac{1}{\eta} \frac{\partial}{\partial c} \sum_{i=1}^{\eta} (y_i - \hat{y}_i)^2$$

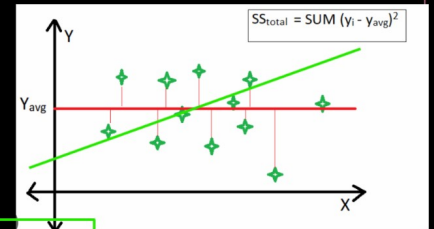
$$\frac{\partial L}{\partial c} = \frac{1}{\eta} \frac{\partial}{\partial c} [y_i - (mx + c)]^2$$

$$\frac{\partial L}{\partial c} = \frac{1}{\eta} * 2 \sum_{i=1}^{\eta} (y_i - (mx_i + c)) * (-1)$$

$$\frac{\partial L}{\partial c} = -\frac{2}{\eta} \sum_{i=1}^{\eta} (y_i - \hat{y}_i)$$

sum of squared for regression

$$R^2 = 1 - \frac{SS_R}{SS_M}$$



Sum of squared

Value for
mean

$$R^2 = 1 -$$

$$\frac{\sum_{i=1}^n \overset{\text{Error}}{(y_i - \hat{y}_i)^2} \downarrow (N)}{\sum_{i=1}^n (y_i - \bar{y}_i)^2 \uparrow (D)}$$

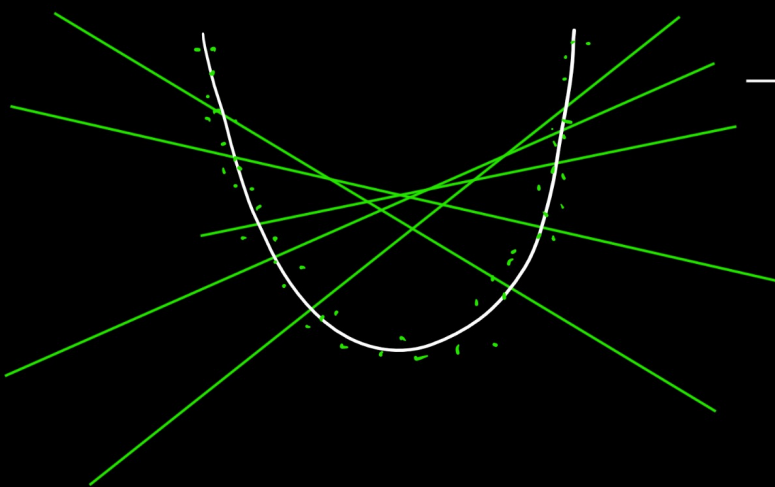
Best fit
Line

$$R^2 \approx 1 - \text{very small value}$$

$$R^2 \approx 1 \text{ (means your model is performing well)}$$

$$R^2 \approx 0 \text{ to } 1 \text{ (Higher the value } \rightarrow \text{ better the results)}$$

$$R^2 \approx 0 \text{ (means your model is performing really bad)}$$



→ Non linear data to Logistic Regression
↳ Linear data